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Aerodynamic and structural analysis for blades of a 15MW floating offshore wind turbine

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ABSTRACT

Recent years have seen an important evolution toward large-scale and increased gross capability in the design of floating wind turbines. For an up-to-date 15 MW wind turbine, the slender blades sustain undesirable risks of failure and fatigue due to the significant interaction between fluid and structure (FSI). Considering its sufficient resolution, a one-way CFD-FEA (Computational Fluid Dynamics-Finite Element Analysis) FSI model is established on a full-scale floating IEA 15 MW Reference Wind Turbine (RWT). Particular attention is focused on the effects of the wave-induced motions on the wind field, the structural response of the blades, and the power generation performance of the floating 15 MW wind turbine. It is found that the relative velocities between the blade and the wind field are evidently disturbed by the pitch and surge motions, resulting in the amplified pressure maxima and abrupt change on the leading edge. Meanwhile, the motions amplify the maximum thrust, torque, and aerodynamic power, but also introduce the minima close to zero. This would lead to significant instability in energy production and possibly a sudden halt of the wind turbine gearbox. Moreover, owing to the detailed 3D composite FEA model, the enlarged stresses and tip deformations of blades due to the platform motions are revealed with better fidelity than the beam model in previous FSI simulations.

1. Introduction

Floating wind turbines have increased exponentially to meet the world's growing renewable energy demand in recent years. To boost the competitiveness of offshore wind energy, the design and construction of floating offshore wind turbines (FOWT) present an important evolution toward large-scale and increased gross capability (Papi and Bianchini, 2022). According to Wind Europe (Ramírez et al., 2021), the average capacity of wind turbines installed in Europe has grown from 6.8 MW in 2018 to 8.2 MW in 2020. Meanwhile, the equipped rotors have been developed in the 12–15 MW range by leading manufacturers like GE Renewable Energy and Vestas (Shields et al., 2021). To scale up the swept area for a declined levelized cost of energy (LCOE), the up-to-date IEA (International Energy Agency) 15 MW offshore wind turbine has a spanwise length of 117 m, approximately twice that of its 5 MW predecessor.

However, the high flexibility and heavy marine loads would introduce undesirable risks and challenging technical requirements to the enlarged wind turbine blades. For one, the flexibility of the slender blades brings many design constraints on the blades' tip deflection loading and tower clearance, which means that the prebend, precone angle, and shaft tilt angle should be selected with more care (Gaertner et al., 2020). For another, the composite structure must be developed to balance the reduction of gross weight and the higher strength demand since the slender blade is one of the locations most prone to failure of a wind turbine (Zheng and Chen, 2022). Moreover, as the floating platform moves in the open sea, the aerodynamic loads on the blades and the structural response would be heavily affected, threatening its normal operations and provoking high risks of failure.

Among the 6-DOF platform motions, the surge and pitch motions are the most influential to the aerodynamic performance and wake characteristics of wind turbines (Tran and Kim, 2016). Fang et al. (2020, 2021) performed computational fluid dynamics simulations to emphasize the effect of the surge and pitch motions on the aerodynamics of a 1:50 model 5 MW FHAWT (Floating Horizontal-Axis Wind Turbine), including the thrust, torque, and rotor power. Chen et al. (2021) investigated the harmonic surge-pitch motion on the aerodynamic characteristics of the NERL 5 MW offshore wind turbine using the unsteady CFD method. It was found that the increase of motion amplitude and frequency aggravated the fluctuation of the overall aerodynamic

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performance of the wind turbine and thus adversely affected its power generation. Nevertheless, the resultant effects of pitch or surge motions on the structural response of wind turbine blades have yet to be figured out thus far. An FSI model is, therefore, crucial to adapting to the fundamental needs for the large-scale development of FOWTs.

Over the past decades, extensive studies on the FSI of wind turbine blades have been conducted, utilizing multiple approaches that vary in accuracy and computational efficiency. The most general approach is to couple BEM (Blade Element Momentum) method with the beam model, which tools of OpenFast, Flex 5, and HWC2 have adopted. Such a coupling approach is highly efficient and suitable for design optimization when a large number of iterations are involved (Y. Liu et al., 2019), while its limitation lies in the fidelity of the structural and aerodynamic components. With respect to the aerodynamic component of FSI modeling, though extensively applied with reasonable accuracy (Pinto and Gonçalves, 2017; Yirtici and Tuncer, 2021), the BEM model is incapable of presenting detailed information on the flow field when it comes to the issues of turbulence, unsteady flow, and wake development (Sanvito et al., 2021). In contrast, the CFD model would accurately simulate complex 3D flow field and visualize realistic fluid dynamics, receiving greater attention in recent years. It has been demonstrated in the unsteady aerodynamic simulation of a horizontal-axis wind turbine (HAWT) by Ye et al. (2020) that the Unsteady Reynolds-Averaged Navier-Stokes (URANS) model had better performance than the BEM model when compared with the experimental results. Guo et al. (2013) coupled CFD with a beam model to acquire the time-domain aeroelastic effects of a large-scale wind turbine in a two-way FSI simulation. Yu and Kwon (2014) coupled the CFD method with a non-linear Euler-Bernoulli beam model to investigate the effect of shaft tilt and tower interference on the aerodynamic performance of a 5 MW RWT (Reference Wind Turbine). Dose et al. (2018) implemented the CFD tool OpenFOAM with isotropic beam codes to examine the effect of blade deformations on wind turbine power, thrust, and sectional forces.

For the structural component of FSI modeling, there have been 1D beam models and 3D FE (Finite Element) models. Some beam models consider cross-sectional properties, for example, a non-linear aeroelastic model for wind turbine blades developed by Wang et al. (2014). Albeit with this, the beam model does not provide further detailed information, which is essential for the designers, such as the stress distribution or deformation within the blade structure. The FE model, however, employs 3D composite shell elements to construct the wind turbine blade and keeps important geometry features, including twist angles, chords, prebend, and pitch angle (Elsherif et al., 2019). These are essential factors that have been emphasized for developing large-scale turbine blades. Boujleben et al. (2020) adopted 3D solid FEA in an FSI simulation to model a full-scale NREL 5 MW RWT with the bending and torsional deflection of the blades well described. More utilization of the FE model in structural analysis can be found in Wiegard et al. (2021), Zhang et al. (2021), and S. Liu et al. (2022).

Coupling CFD and FEA as an undoubtedly high-resolution FSI approach better captures wind turbine blades' aerodynamic effects with detailed structural responses. In particular, the one-way coupling for the CFD-FEA data transmission is capable of obtaining the aerodynamic and flow characteristics in a more efficient way (Elsherif et al., 2019; Lee et al., 2017; Roul & Kumar, 2020; Wang et al., 2016), compared with the two-way coupling approach. Its sufficient resolution has been verified by the comparative studies of Borouji and Nishino (2019) and Sayed et al. (2019), which reported that the thrust coefficient of 5 MW RWT and 10 MW RWT is only 3.76% and 1% smaller in one-way simulation than that in two-way simulation. The pressure distribution revealed in Borouji and Nishino (2019) showed no obvious distinction between one-way and two-way cases. Likewise, the study of Carrión et al. (2014) revealed that the differences in the integrated thrust and torque between the rigid and the elastic blades were almost negligible.

Hence, in this study, a CFD-FEA one-way coupling simulation is performed to investigate the effects of the wave-induced motions on the wind field, structural response of the blades and the power generation performance of a floating 15 MW wind turbine, which has not been previously studied in a detailed manner. To improve the fidelity of the structural and aerodynamic components, a detailed FE model with 3D composite shell elements is employed to construct the full-scale wind turbine blade. It is found that the relative velocities between the blade and the wind field are evidently disturbed by the FOWT motion, resulting in amplified fluctuations in pressure distribution as well as the performance values. Owing to the detailed 3D composite FE model, the amplification of maximum stress and tip deformations as well as locations prone to failure are provided accurately. This paper is structured as follows: Section 2 introduces the FSI methodology, including the wind turbine model, CFD modeling, FEA modeling, and one-way mapping. Results and discussions are given in Section 3, followed by the conclusion summarized in Section 4.

2. FOWT model and numerical methods

2.1. Wind turbine model

The subject of the numerical model is based on the 15 MW horizontal-axis offshore wind turbine (Gaertner et al., 2020), the platform of which is the VolturnUS-S Semi-Submersible (Allen et al., 2020). This newly developed wind turbine consists of long, slender blades to acquire substantial power while effectively reducing the structural weight. With three 117-m blades, the horizontal-axis wind turbine has a rotor diameter of 240 m and a rated wind speed of 10.59 m/s. Detailed parameters of the 15 MW wind turbine are shown in Table 1. In this study, the IEA 15 MW wind turbine rotor is simulated at full scale. Fig. 1 provides the geometry model of the rotor and blades established in the numerical simulation. From root to tip, the blade cross-sectional geometries are designed by DTU FFA-W3 airfoils, whose relative thicknesses vary from 0.5 to 0.211, as shown in Fig. 2 (a). To settle the blade geometry accurately, 53 sets of airfoil data with exact chord length, twist angle, and prebend distance (illustrated in Fig. 2 (b)) is applied in the simulation based on the wind turbine ontology developed by Bortolotti et al. (2022). Additionally, the 4-m prebend and 4-degree precone have been taken into consideration when modeling the rotor.

To afford both wave impact and wind load, the rotating plane of the FOWT is constrained in six degrees of freedom (6-DOFs). As illustrated in Fig. 3, the three translation components are surge along the x-axis, sway along the y-axis and heave along the z-axis, and the corresponding rotations are roll, pitch, and yaw. Specifically, in this study, the emphasis is focused on the surge and pitch motions, and the effects on the wind turbines' aerodynamic performance and structural response will be examined. Limited by the efficiency of the calculation, only one set of period and amplitude is chosen for each motion according to the reported literature such as (Fang et al., 2020, 2021). We hope to conduct more numerical tests with different periods and amplitudes of the FOWT motion in the future and perform a deeper study on their influence on the structural characteristics of the 15 MW RWT blade.

The surge and pitch motions of the floating IEA 15 MW wind turbine

Table 1
Key parameters of the IEA 15 MW RWT.

Parameter	Value	Unit
Power rating	15	MW
Rated wind speed	10.59	m/s
Design tip-speed ratio	9.0	–
Rotor diameter	240	m
Rotor precone angle	−4.0	deg
Shaft tilt angle	6	deg
Blade length	117	m
Blade prebend	4	m
Hub overhang	11.35	m
Hub height	150	m

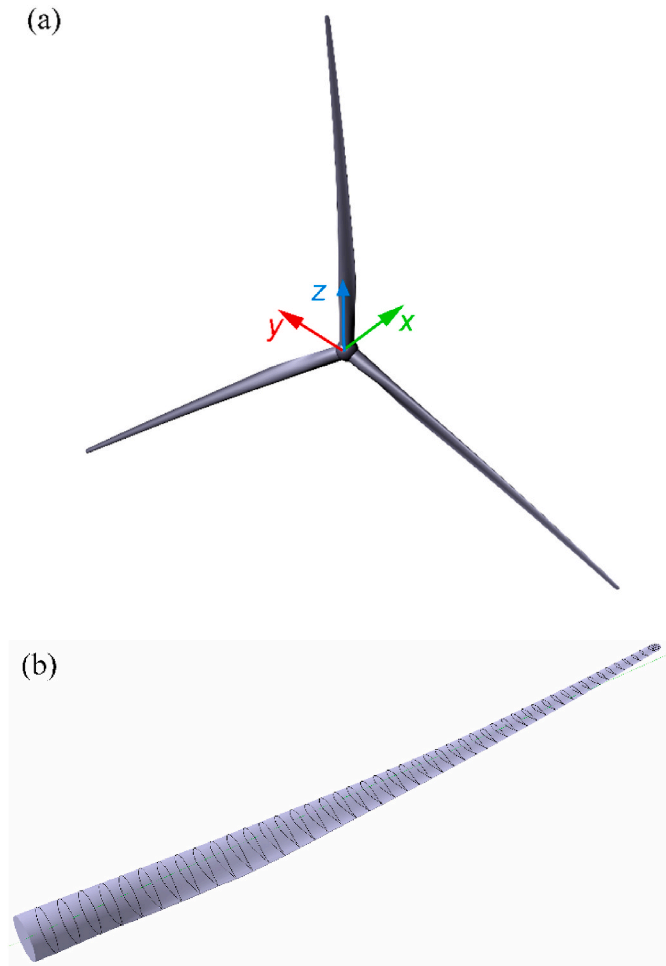

Fig. 1. 3D geometry model of the rotor and blades of IEA 15 MW wind turbine (a) Rotor; (b) Single blade.

are considered simple harmonic motions. The motions with specific periods and amplitudes are defined, in which the rotation of the rotor is set as a superimposed motion with the constant rotational angular speed. The surge motion displacement is defined as

$$D_{surge} = a \sin\left(\frac{2\pi}{T}t\right) \tag{1}$$

The velocity of surge motion is the first derivative of the displacement, and the acceleration is the second. Therefore, they can be described as

$$V_{surge} = \left(\frac{2\pi a}{T}\right)\cos\left(\frac{2\pi}{T}t\right) \tag{2}$$

and

$$A_{surge} = -\left(\frac{4\pi^2 a}{T^2}\right)\sin\left(\frac{2\pi}{T}t\right) \tag{3}$$

where T and a represent the oscillation period and the surge amplitude, respectively, which are set as 4s and 4 m. The direction of the surge motion is along the rotation axis of the rotor.

The pitch center in this simulation is presumed to be at the root of the tower. The angular displacement, velocity and acceleration of the pitch motion are represented as

$$\theta_{pitch} = \alpha \sin\left(\frac{2\pi}{T}t\right) \tag{4}$$

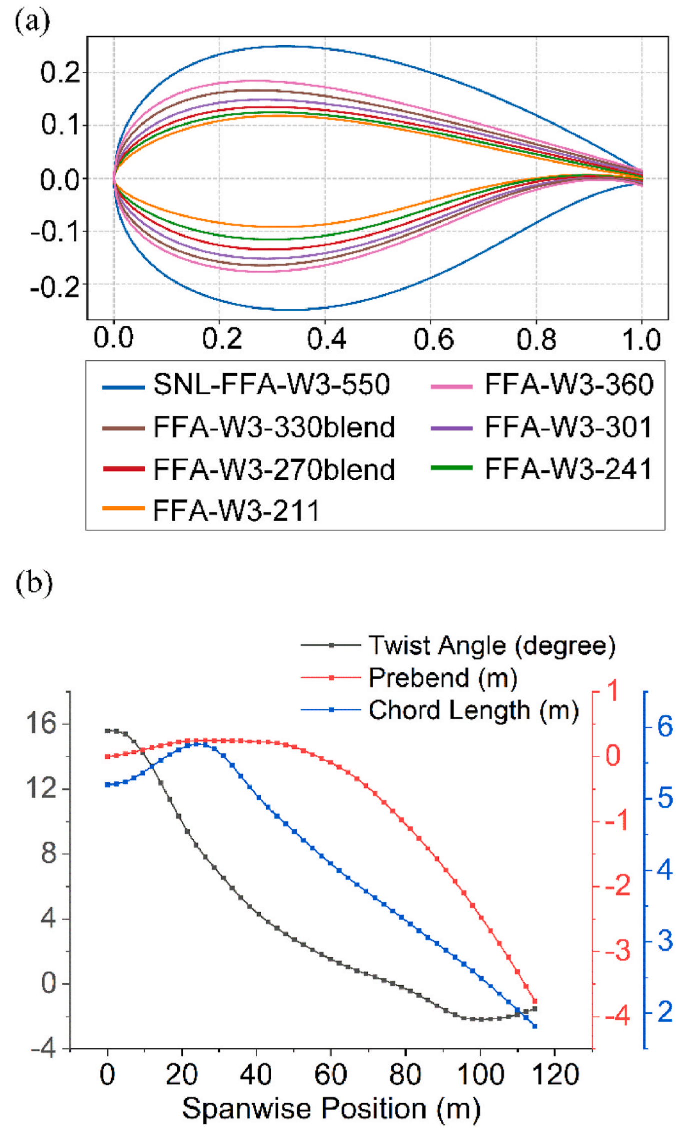

Fig. 2. Airfoils and geometrical parameters of the 15 MW wind turbine blade (a) DTU FFA-W3 airfoils (Gaertner et al., 2020); (b) Twist angle, prebend, and chord length varying with the spanwise position.

$$\omega_{pitch} = \left(\frac{2\pi\alpha}{T}\right)\cos\left(\frac{2\pi}{T}t\right) \tag{5}$$

and

$$A_{pitch} = -\left(\frac{4\pi^2\alpha}{T^2}\right)\sin\left(\frac{2\pi}{T}t\right) \tag{6}$$

where T represents the oscillation period and α the pitch amplitude, which are set as 4s and 2 deg, respectively.

2.2. Flow solver

To acquire the aerodynamic pressure and simulate the aerodynamic performance of the IEA 15 MW wind turbine, the CFD model of the wind turbine rotor is established and solved using the software STAR-CCM+ (Siemens, 2021). Detailed information related to the simulation, including the selected turbulence model and solution method, the setting of the computational domain, the mesh topology, and the resolution analysis is presented hereinafter.

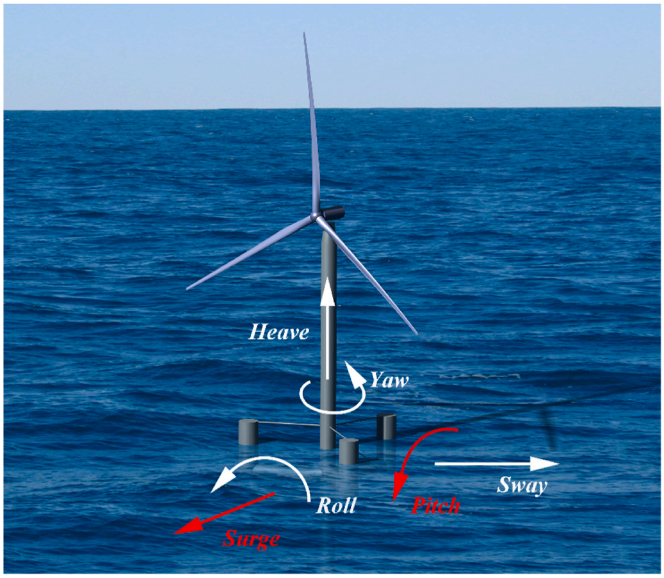

Fig. 3. Six-DOF motions of the IEA 15 MW wind turbine.

2.2.1. CFD model and solution method

The solution for the flow field is achieved based on solving the incompressible URANS equations. The closure problem is treated by the Realizable k-ε turbulence model by Shih et al. (1994). This two-equation eddy viscosity model has been proven to have an excellent performance for complex turbulent flows, including rotating flow, flow separation, and boundary layer flow with strong adverse inverse pressure gradient (Hashem and Mohamed, 2018). Hence, this model has been widely used in the aerodynamics of wind turbines with favorable results, such as Hu and Ding (2016), Borouji and Nishino (2019), Kaya et al. (2020), and Zhang et al. (2021).

The governing equations are discretized in the numerical calculation with the Finite Volume Method (FVM). With designated time steps, time discretization follows the Implicit Euler Method, approximating the transient term using data from the current and the previous moments. Such is a method of first-order accuracy. A SIMPLE algorithm is selected for the pressure-velocity coupling. The second-order upwind scheme is adopted for the dispersion of pressure, momentum, turbulent kinetic energy, and dissipation rate. The convergence of residuals is calibrated and examined to ensure sufficient accuracy. The residual values of the continuous equation, momentum equation, and energy equation are set as $o\ (10^{-3})$.

2.2.2. Computational domain and mesh topology

The computational domain and boundary conditions for the CFD model are depicted in Fig. 4. As shown in Fig. 4, the whole computational domain is composed of a cylindrical volume background domain and an overset domain containing the rotor. The external cylindrical domain is a stationary zone of 9D in length and 6D in radius, where D is the rotor diameter. The overset domain is set for the rotor with a radius of 1.17D and 0.17D in length. As is shown in Fig. 5 (a), three refined domains are designed in the background domain to improve accuracy and cost control. The first refined domain has the length of the whole domain and a radius of 3D for reducing the numerical diffusion of the inlet wind. The second is designed to increase the grid density for better resolution of wake flow, with the size of 1.5D × 4.5D. The third refined domain is designed explicitly for the surge and pitch motions, which is large enough to surround the overset domain during the motions. It is applied with a radius of 1.33D and extended 0.75D in upstream and downstream directions. To reveal the flow details around the blades, in the overset domain, three combined conical volumes are set to enclose the rotor with refined mesh. To meet the requirements of $y+$ values,

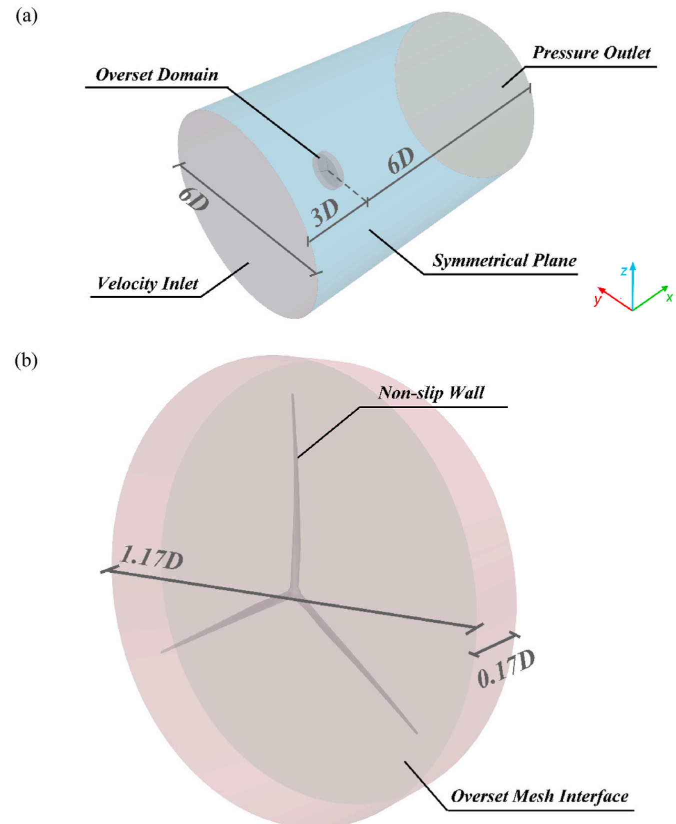

Fig. 4. Computational domain and boundary conditions for CFD modeling.

separate parts enclosing the leading and trailing edges are modified with finer grids.

Based on the right-handed Cartesian coordinate system, the origin of the coordinate system is established on the projection of the center of the wind turbine rotor. As shown in Fig. 4 (a), the positive direction of the x-axis is the wind flow direction, the y-axis is perpendicular to the wind direction, and the z-axis is vertically upward. The left boundary of the background domain is defined as the velocity inlet and the right end is defined as the pressure outlet. The side wall of the domain is defined as the symmetrical plane. The surface of the wind turbine blades, which is related to the aerodynamic characteristics and subject to aerodynamic pressure, is regarded as a stationary non-slip wall. Specifically, the interface between the background and overset domain is set as overset mesh interface. In the simulation, the wind speed is considered as the rated speed of 10.569 m/s, which means the rotor rotates at 7.4992 rpm, and this is set in the domain inlet as well as in the initial conditions.

2.2.3. Spatial and temporal resolution study

A spatial and temporal resolution study is carried out between three mesh cases and three time-step cases under the rated wind speed. The mesh topology of the CFD model is presented in Fig. 5. In the spatial resolution study, the unstructured trimmed mesh is applied to the background and overset domains. The base size of the background domain has been set as 50 m, 40 m, and 32 m, respectively, for the coarse, intermediate, and fine mesh cases. The refinement strategy for the three cases is identical. Take the intermediate case, for example. The mesh size is cut in half successively for the first, second, and third refined domains, as shown in Fig. 5 (a). In this way, the mesh sizes around and inside the overset domain are both 5 m, and the three-cone volume in the overset domain limits the mesh size around the blades to 2.5 m. Moreover, another mesh control is applied on the rotor surface to define the size of the surface mesh and its adjacent volume mesh as 0.2 m. For a more accurate solution of the near-wall flow, a prism layer mesh

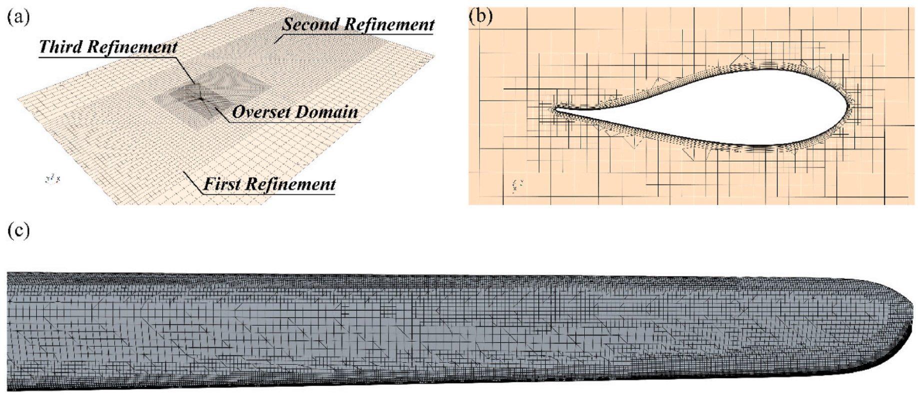

Fig. 5. Mesh topology of the CFD model.

type is adopted in the near-wall regions, and 15 layers of prism layer meshes are set with a ratio of 1.3, which is presented in Fig. 5 (b). The thinnest first prism layer that appears at the trailing edge is 8×10^{-5} m, and its adjacent volume mesh size is 7.8125×10^{-2} m. The mesh around the blade tip is shown in Fig. 5 (c). A high $y+$ treatment is adopted in this study as the $y+$ value is within the range of 0∼200. The simulated thrust and torque values of the three mesh cases are compared with the designed performance values in Table 2. Considering the computational expense and the precision of the simulation results, the intermediate case is selected with satisfactory errors −4.573% in thrust and −0.359% in torque. The errors are the ratio of the difference between calculated and designed performance values to the performance values. Further investigation on the surging and pitching rotors is also based on this 22-million mesh case.

In the temporal resolution study, the time steps equal the time required for the rotor to rotate 0.5°, 1°, and 2°, respectively. All three cases have adopted the same mesh strategy as the intermediate mesh case. As shown in Table 3, more minor time step cases give results more aligned with the design performance values. Also, cases with smaller time steps converge faster. However, it takes 10,800 iterations for a 0.5°-time-step case to finish a 24s simulation, indicating an excessive calculation that contributes little to precision. The one-degree time step is thus chosen, which is 0.0224s, considering both the efficiency and accuracy of the simulation.

Table 3
CFD temporal resolution study.

Cases	Base size (m)	Iteration (Times)	Thrust		Torque	
			Simulated (MN)	Relative error (%)	Simulated (MN)	Relative error (%)
0.5°	40	10,800	2.3285	−4.477	19.913	−0.171
1°	40	5400	2.3261	−4.573	19.875	−0.359
2°	40	2700	2.3258	−4.587	19.857	−0.454

2.3. Structure solver

2.3.1. FEA modeling

A high-fidelity FE model of the wind turbine blade is developed in MSC Nastran, aided by the pre-processing code HyperMesh and post-processing code HyperView. The composite structure of the blade is modeled using PCOMP shells, including the outer shell, two shear webs, and the spar caps. Fig. 6 gives the schematic of the blade structure modeled in this FEA simulation. The shear webs (Fig. 6 (b)), which deviate 350 mm from the blade spanwise central plane, are presented as supporting plates extending from the blade root to the tip. The spar caps (Fig. 6 (c)) are located on the suction and pressure sides of the blade to provide additional stiffness. The hub is not included in this model since it is assumed to be a rigid body.

The 15 MW wind turbine blade is made of five types of materials, and the property parameters are listed in Table 4. The outer surface of the blade shell is enclosed by an extra ply of Gelcoat material, forming a UV Protection layer for the wind turbine blade. The spar caps are filled with the material CarbonUD, while the rest parts of the blade shell and the shear webs are stuffed with medium density foam. For shell skin and web skin, the Glass Triax and Glass Biax materials are utilized, respectively. In this modeling of the composite layers, the material orientations of all elements are set in the spanwise direction. The thickness of each ply varying with spanwise position is depicted in Fig. 7. It can be seen that the thickness of the blade shell and shear webs reduce gradually from the root to the tip. In contrast, that of the blade shell with spar caps increases steadily to a maximum thickness above 90 mm at the spanwise direction of 28–56% before a sharp decrease. This would effectively resist bending moment and reduce deformation of the flexible blades.

2.3.2. FEA mesh strategy and solution setting

Applying HyperMesh, the blade structure is meshed using a 2D mixed mesh type. This is selected to capture detailed features of the complex blade surface, which crucially concerns the aerodynamic characteristics of the wind turbine. A grid convergence study is conducted to determine an appropriate mesh size, including four cases, from Case A (0.10–0.25) to D (0.04–0.10). The case names contain each mesh scheme's maximum and minimum mesh sizes, which appear at the root and the tip, respectively. For example, the element size of Case A is set as 0.25 m at the blade's root and is decreased gradually to 0.10 m at the tip. The results of the different mesh cases are compared for the first five modal frequencies in Table 5. It is seen that the frequencies generally increase as the mesh grows finer but at minor rates. The modal frequencies of Case D (0.10–0.04) grow 0.42%, 0.04%, 0.01%, 0.05%, and

Table 2
CFD spatial resolution study.

Cases	Time step	Mesh			Thrust		Torque	
		Base size (m)	Background	Overset	Simulated (MN)	Relative error (%)	Simulated (MN)	Relative error (%)
Coarse	1°	50	182,440	19,781,011	2.311	−5.183	19.439	−2.545
Intermediate	1°	40	347,844	22,320,687	2.326	−4.573	19.875	−0.359
Fine	1°	32	658,928	29,837,671	2.327	−4.554	19.981	0.172

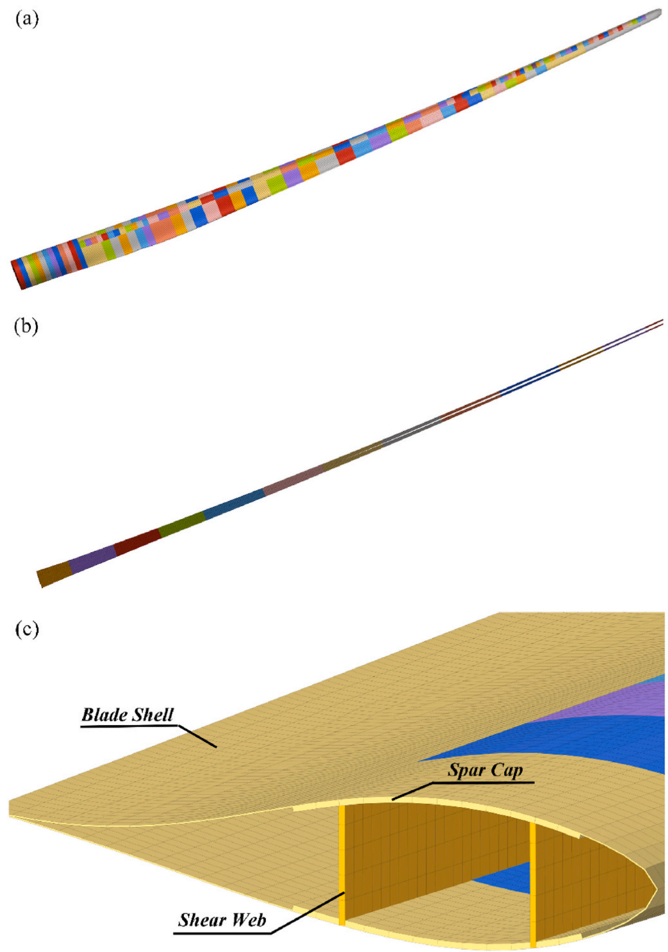

Fig. 6. Schematic of the blade structure modeled in the FEA simulation (a) Blade shell; (b) Shear webs; (c) Internal structure.

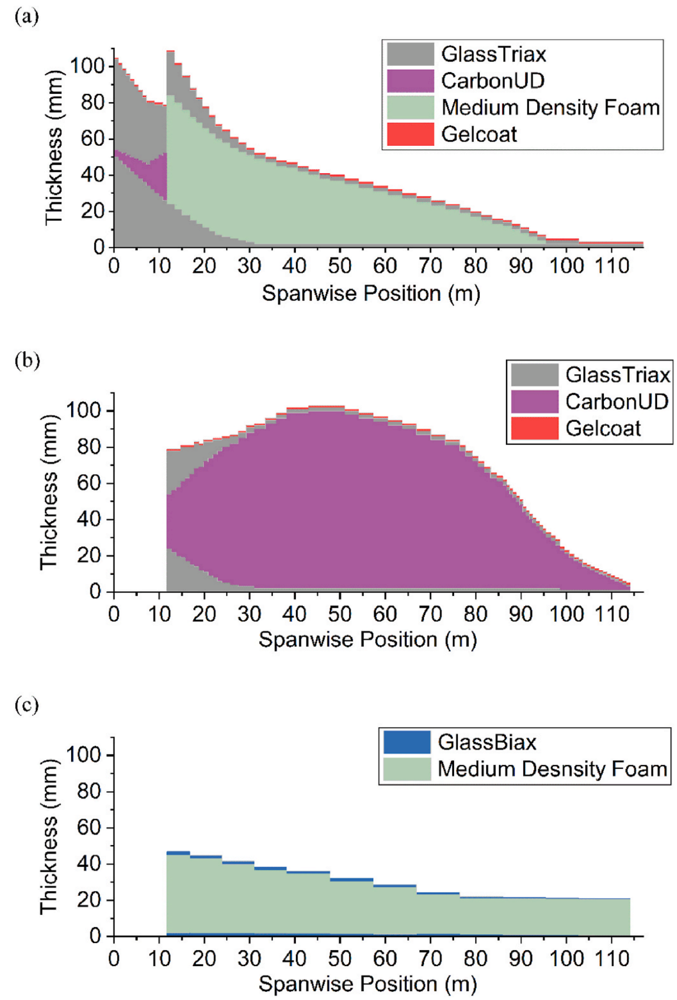

Fig. 7. Thickness of each ply varying along the span in the FE model (a) Blade shell; (b) Blade shell with spar caps; (c) Shear webs.

−0.14%, respectively, over those of Case C (0.15–0.06), demonstrating a good convergence. Thus, Case C is deemed a proper mesh scheme and utilized in this study, considering both the fidelity and expense of the calculation.

Since the hub of the wind turbine is assumed to be a rigid body, the roots of the wind turbine blades are considered fixed, and the six DOFs of all the nodes in the blades' roots are constrained. In such a case, the constraints of a wind turbine blade resemble those of a cantilever. It is worth noting that to account for the influence of inertial loads, the acceleration of the surge and the rotational acceleration and velocity of the pitch motion are applied to the blade structure, as depicted in Eqs. (3), (5) and (6). The gravitational force is also applied. The aerodynamic pressure acquired from the CFD simulation will be imported and set as the load on the outer surface blade shell.

2.4. Coupling approach and mapping procedure

As mentioned earlier, the one-way fluid-structure coupling model is applied in this study, considering its relatively low computational cost and sufficient resolution. In the one-way coupling, the surface pressure of the loads added to the blade surface in FEA modeling is acquired from the aerodynamic pressure data by CFD simulations.

In this study, the partitioned one-way coupling is realized by data mapping using *bdf*-format data files. The aerodynamic pressure acquired from CFD simulations is discretized and mapped to each mesh element of the blade's positive surface in the FE model. In this process, the blade boundaries in the CFD model are set as the source surface. Hence, the source data is face-centric, while the target points are the centers of the shell elements (Siemens, 2021). As a form of mathematical regression analysis, the Least Squares Method is used here to interpolate data between the CFD model's rotor surface and the FE model's outer blade shell. To calculate the mapping value at the target point *a*, the first-order Taylor series is used to approximate the solution field *f*, taking point *an* as the center, written as

Table 4
Materials in the FE model.

Material	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	ρ (t/mm^3)
Gelcoat	3.440×10^3	3.440×10^3	1.323×10^3	0.300	1.235×10^{-9}
Medium Density Foam	1.292×10^2	1.292×10^2	4.895×10^1	0.320	1.300×10^{-10}
CarbonUD	1.145×10^5	8.390×10^3	5.990×10^3	0.270	1.220×10^{-9}
Glass Triax	2.870×10^4	1.660×10^4	8.400×10^3	0.500	1.940×10^{-9}
Glass Biax	1.110×10^4	1.110×10^4	1.353×10^4	0.500	1.940×10^{-9}

Table 5
FEA mesh sensitivity test.

Case	1st Modal Frequency	2nd Modal Frequency	3rd Modal Frequency	4th Modal Frequency	5th Modal Frequency	Number of Elements
A (0.25–0.10)	0.46431	0.54676	1.4528	1.6711	2.8345	33,952
B (0.20–0.08)	0.46451	0.54706	1.4537	1.6718	2.8366	52,831
C (0.15–0.06)	0.46619	0.54745	1.4546	1.6748	2.8375	93,838
D (0.10–0.04)	0.46815	0.54769	1.4547	1.6757	2.8336	211,397

$$f(\mathbf{x}) \approx f(\mathbf{x}_a) + (\mathbf{x} - \mathbf{x}_a)\nabla f(\mathbf{x}_a) \tag{7}$$

Following the least squares method, $f(\mathbf{x}_a)$ is calculated by minimizing the sum of squared residuals, and the residual is the difference between the predicted value $f(\mathbf{x}_i)$ at the source location and its known value f_i

$$S = \sum_{i \in F(c)} r_i^2 \tag{8}$$

$$r_i = f(\mathbf{x}_i) - f_i \tag{9}$$

$F(c)$ denotes the centroid group of the participating surfaces, which are adjacent to and sharing at least one node with the surface closest to the target point. After the convergence of data transmission, the *bdf*-format file that contains the extra aerodynamic pressure is imported into Nastran to carry on an FEA analysis.

To verify the accuracy and stability of this process, the distribution of CFD-calculated and the mapped pressure is compared in Fig. 8. Two different FEA element types are examined: the mixed type (mainly quad meshes, used in this study) and the tri type (all tri meshes). The mesh sizes are both set to be 150 mm. It can be found that the pressure distribution of the FE models is consistent with that of the CFD model, demonstrating that the mapping process is successful. With an isotropic material property ($E = 7.2 \times 10^4$, $\rho = 1.8 \times 10^{-9}$, $\nu = 0.2$, $thickness = 20$ mm) assigned, the two FE models give quite close results. The tip deformation for the mix-element and tri-element models are 38.4 m and 38.5 m, respectively. This verifies that although with distinct target elements, the pressure is mapped accurately, and the results given by the FE analysis are stable.

2.5. Validation

2.5.1. Validation for the CFD model

The performance parameters of the wind turbine, including the power, torque, and thrust, are usually regarded as the benchmark parameters of a wind turbine's working state. To validate the results acquired from the CFD simulation, the parameters simulated in this study are compared with the design performance listed in the official file (IEA Wind Task 37 Team, 2022/2022). The targeted steady-state performance is calculated using OpenFast with the ROSCO controller. While in this study, the aerodynamic performance parameters are derived from the simulated aerodynamic pressure on the rotor surface using field functions (Siemens, 2021).

The torque about the rotational axis of the rotor is defined as

$$M = \sum_f \left[\overrightarrow{r_f} \times \left(\overrightarrow{f_f^{pressure}} + \overrightarrow{f_f^{shear}} \right) \right] \cdot \overrightarrow{a} \tag{10}$$

where $\overrightarrow{f_f^{pressure}}$ and $\overrightarrow{f_f^{shear}}$ are the pressure and shear force vectors, $\overrightarrow{a}$ is a vector defining the rotational axis about which the toque is taken, and $\overrightarrow{r_f}$ is the position of face with f relative to the origin.

The thrust is computed as

$$T = \sum_f \rho_f \overrightarrow{v_f} \overrightarrow{v_f} \cdot \overrightarrow{a_f} + \sum_f (p_f - p_{atm}) \cdot \overrightarrow{a_f} \tag{11}$$

where ρ_f is the fluid density, $\overrightarrow{v_f}$ is the velocity vector, p_f is the exit pressure, p_{atm} is the atmospheric pressure, and $\overrightarrow{a_f}$ is the surface area vector of the face.

The power equals the product of torque and rotational velocity, written as

$$P = M \cdot \left(\frac{2\pi\omega}{60} \right) \tag{12}$$

where ω is the rotational velocity with the unit of rpm.

The results of the thrust, torque, and power under rated wind speed are obtained and presented in Fig. 9. The values fluctuate at a meager rate after 16 s of simulation. The relative errors in the intermediate case column of Table 2 show good agreements between the targeted and simulated values. This demonstrates the feasibility of the current URANS model in investigating the aerodynamics of the IEA 15 MW wind turbine.

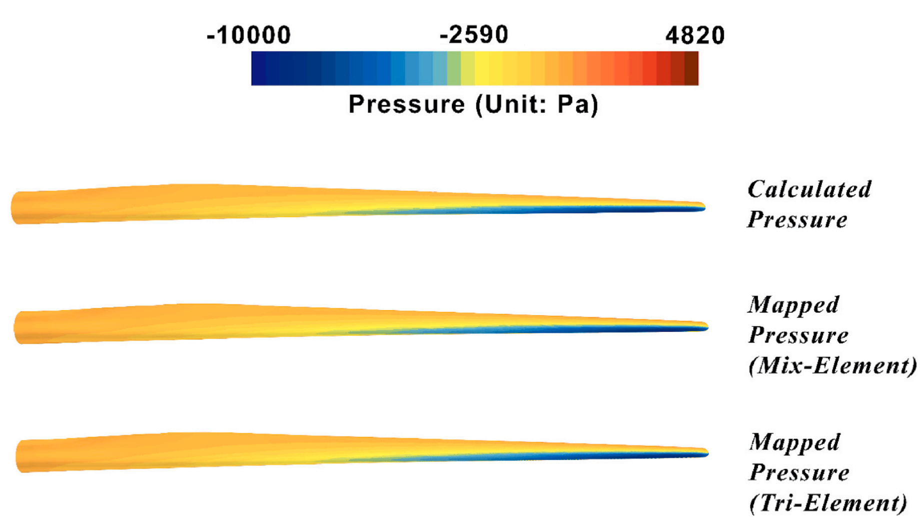

Fig. 8. Comparison of calculated and mapped pressure distribution on the blade's suction side.

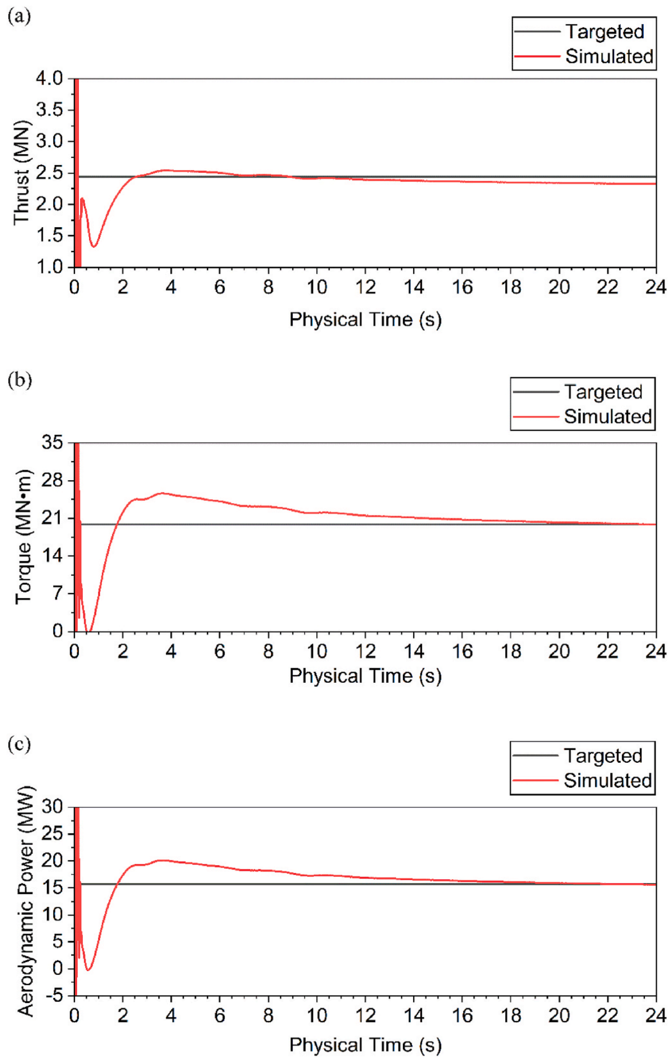

Fig. 9. Rotor performance values over the physical time (a) Torque; (b) Thrust; (c) Aerodynamic Power.

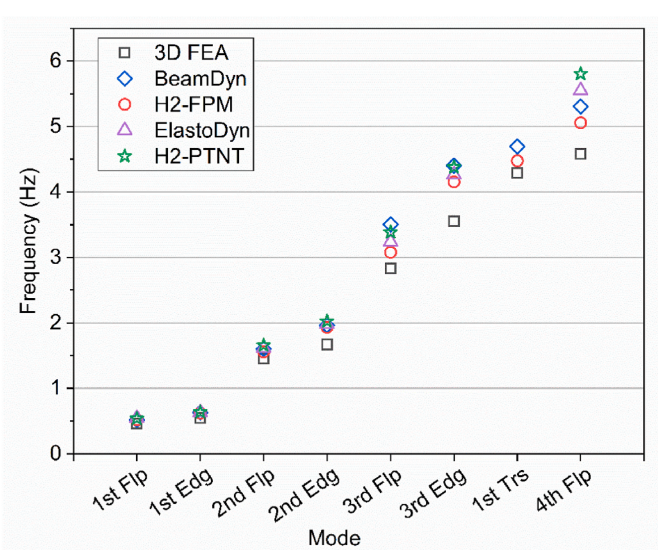

Fig. 10. Modal frequencies of the IEA 15 MW RWT blade.

2.5.2. Validation for the FE model

The FE model is validated by comparing the modal frequencies with those acquired from the beam model. In this validation, four different models are selected from the study of Rinker et al. (2020), including ElastoDyn, BeamDyn, prismatic Timoshenko without torsion (H2-PTNT), and Timoshenko with fully populated stiffness matrix (H2-FPM). Fig. 10 shows the first eight modal frequencies of the IEA 15 MW RWT blade, calculated by the 3D FE model of this study and the four beam models, and the detailed data are presented in Table 6. As is shown in Fig. 10, there are three types of blade modes, which are flapwise, edgewise, and torsional modes. The modal frequencies of the FE model are close to those of the beam models, especially for the first three modes, and they are generally smaller. In particular, frequencies of BeamDyn and H2-FVM models are numerically closer to the 3D FE model. This is because the BeamDyn and H2-FPM models of the four possess higher fidelity, modeled with additional stiffness matrices and taking into account the torsional deformations. While torsions are not considered in ElastoDyn and H2-PTNT models, they do not possess the first torsional frequency. Hence, the validity of the FE model is confirmed.

3. Results and discussion

Given the rated wind speed, cases of the IEA 15 MW wind turbine rotor rotating with pitch or surge motions and without motions (named "fixed" hereinafter) have been simulated with different CFD models. Afterward, the acquired aerodynamic pressure is mapped to the FE model to examine the blades' deformation and stress distribution. The results from CFD and FE models are presented in the following sections 3.1 and 3.2.

3.1. Aerodynamic characteristics

3.1.1. Flow pattern

The CFD simulation calculates detailed flow patterns as the rotor moves and rotates within the wind field. To reduce calculation time, the simple-harmonic motions are set to occur at 24s when the benchmark performance values fully converge. Compared with the wind velocities around the fixed blade, Fig. 11 gives wind velocities around the blade under pitch and surge motion at the spanwise position of $r/R = 0.8$ within one period. It is clearly observed that the harmonic excitation directly influences the relative velocity between the rotor and the wind, leading to the periodic characteristics in the relative velocity around blades under pitch and surge motions. As illustrated in Fig. 11, the wind speed on the suction surface in all situations of fixed and surge cases and $0.25T{\sim}1T$ of the pitch case is larger than that on the pressure surface, generating the lift force on the blade. Compared with the wind velocity of the fixed wind turbine around the $r/R = 0.8$ profile, the wind velocity for the wind turbine of surge motion is significantly intensified for the moment $0.5T$, while it is weakened relatively for the moments of $0T$ and $1T$. Similar tendencies are found in the wind velocities adjacent to the wind turbine with pitch motion. Specifically, the wind velocity field of the pitch case presents different patterns for the moments of $0T$ and $1T$, which is related to the variation of relative velocities between the wind and blades. The relative velocities strongly depend on the blade's position and vary with the distance between the profile and the pitch center. At $0T$, which in this case is $t = 32$s, the blade from which the profile is abstracted is located at the 0°-azimuth position, while at $1T$, in physical time $t = 36$s, it is located at 180°-azimuth position. The blade tips at these two positions are the top-most and bottom-most points of the rotor. The local pitch velocity on the tip, calculated from Eq. (5), is pointing in the same direction as wind velocity but exceeds its value at $0T$. Hence, the relative speed of the blade tip is −12.21 m/s, while at $1T$, the relative speed of the blade tip is 7.53 m/s. The negative velocity alters the wind velocity field in the pitch case at $0T$, as is shown in Fig. 11, the wind velocity on the pressure surface is larger than that on

Table 6
Modal frequencies of the IEA 15 MW RWT blade (unit: Hz).

Model	1st flapwise	1st edgewise	2nd flapwise	2nd edgewise	3rd flapwise	3rd edgewise	1st torsional	4th flapwise
3D FE	0.465	0.547	1.452	1.671	2.834	3.554	4.295	4.580
H2-FPM	0.521	0.619	1.559	1.933	3.078	4.156	4.475	5.506
BeamDyn	0.517	0.630	1.603	1.964	3.506	4.403	4.694	5.307
H2-PTNT	0.538	0.638	1.658	2.024	3.385	4.381	–	5.805
ElastoDyn	0.545	0.631	1.618	1.981	3.239	4.264	–	5.552

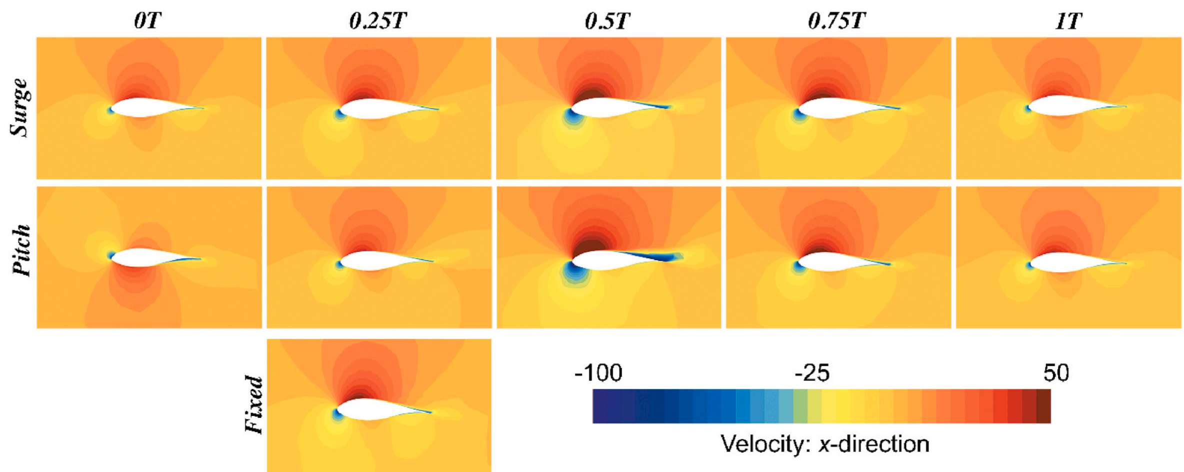

Fig. 11. Wind velocities around the blade under pitch and surge motions and without motions at the spanwise position of $r/R = 0.8$ within one period.

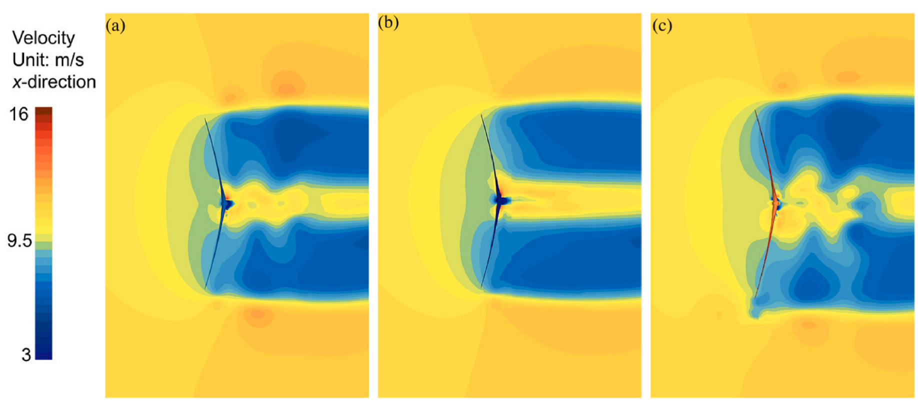

Fig. 12. Far field velocities of the computational domain in the x-direction for surge, pitch, and fixed cases (a) Surge; (b) Fixed; (c) Pitch.

the suction surface, which is distinctive from other situations.

The CFD simulation also captures specific changes in the wake flow as the rotor pitches or surges. Fig. 12 illustrates the far field velocities of the computational domain in the x-direction for surge, pitch, and fixed cases of the offshore wind turbine. Flow separation is seen on the suction side of the rotor in Fig. 12, which is also revealed in Fig. R18 in (Borouji and Nishino, 2019). It is evident in Fig. 12 that the flow field is almost symmetrical for a fixed offshore wind turbine. In contrast, in the cases of the surging wind turbine, the vortex spreads towards both sides of the z-direction. The high-velocity region behind the hub expands, and the low-velocity region is also disturbed by increased speed. Concerning the wind turbine with pitch motion, it is also seen a distortion of low-velocity regions and an expansion of high-velocity regions, except that the asymmetricity is considerably aggravated, with the lower part of the low-velocity region extruding and the vortexes spreading more violently.

3.1.2. Pressure distribution

The pressure distribution of the rotor is acquired from the CFD simulation. With the 4-s-period pitch or surge motions, the surface pressure of the rotor in the two cases oscillates in the same period, compared with that of the fixed rotating case. The pressure distribution of the pitch and surge at the spanwise position of $r/R = 0.5$–0.95 within one period are compared with those of the fixed cases in the form of pressure coefficient in Figs. 13–14 and pressure contour in Fig. 15. The pressure coefficient is defined as

$$C_p = \frac{p - p_0}{0.5\rho(\omega r)^2} \tag{13}$$

where ρ is the air density; p is the absolute pressure; p_0 is the standard atmospheric pressure; ω is the rotational velocity of the rotor; and r is the distance between a profile and the rotational center.

It is evident in Figs. 13–14 that the blade's leading edge ($x/c = 0$) sustains the largest positive and negative pressure. As it draws close to the trailing edge ($x/c = 1$), the pressure on the pressure surface

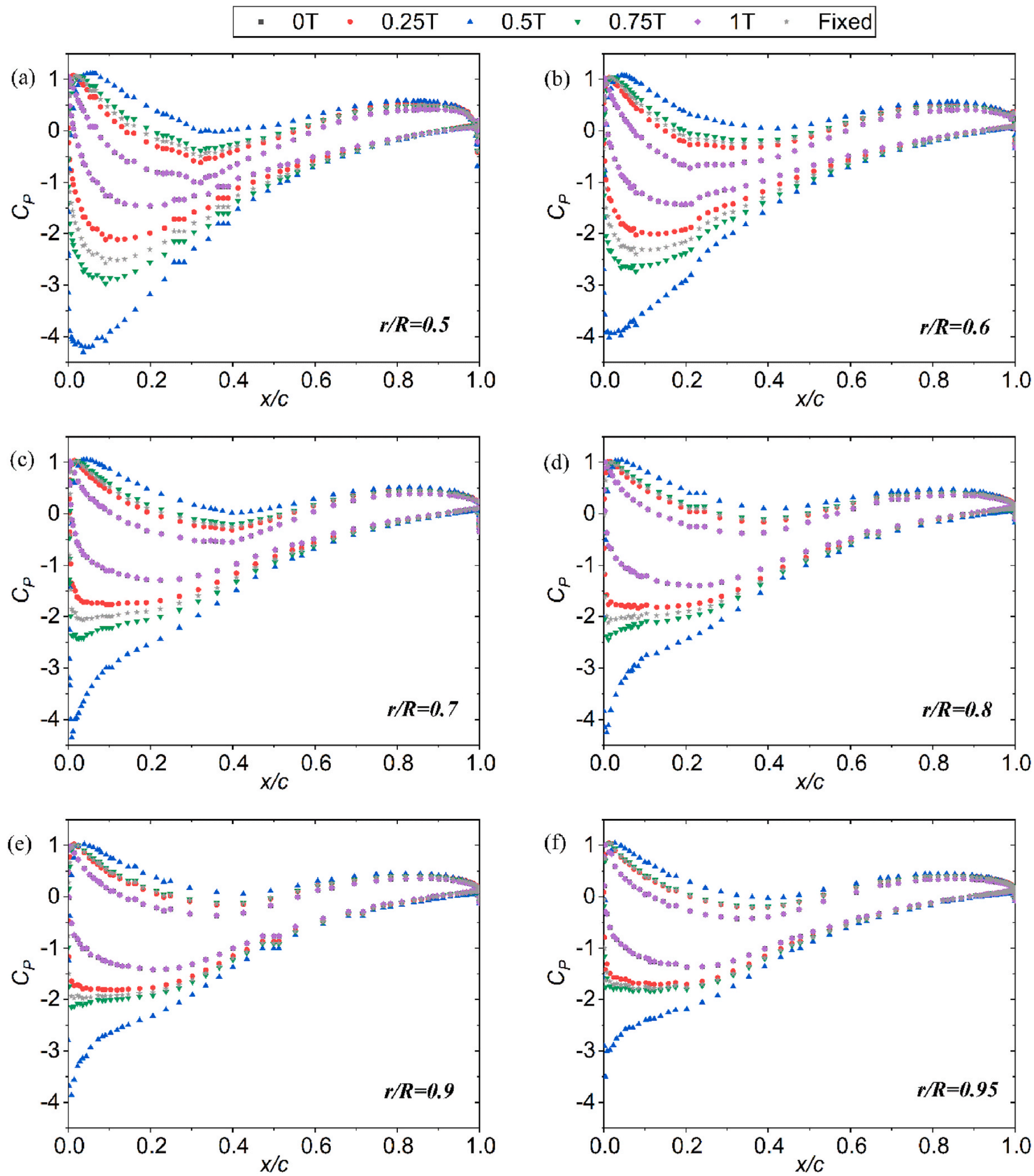

Fig. 13. Pressure coefficients under surge motion at the spanwise position of $r/R = 0.5$–0.95 within one period (a) $r/R = 0.5$ (b) $r/R = 0.6$ (c) $r/R = 0.7$ (d) $r/R = 0.8$ (e) $r/R = 0.9$ (f) $r/R = 0.95$.

decreases sharply and then slightly increases, while that on the suction surface decreases. Specifically, the positive pressure is lowest at positions of $x/c = 0.2$–0.5, and the minimum negative pressure appears on the trailing edge. Also, the pressure distribution varies across one period (T). The pressure coefficients hit the peak at $0.5T$ as the rotor passes through the original position with the highest relative velocity. For the surge case in Fig. 13, the pressure coefficients at $0.0T$ and $1.0T$, which approximately overlap with each other, are relatively small due to the lowest values of the surge velocities. In contrast, for the scenario with pitch motion in Fig. 14, both the positive and negative pressure coefficients are significantly higher at $0T$ than $1T$ for $r/R = 0.50$–0.60. With higher velocities, the maximum pressure in the pitch case is considerably larger than that of the surge case, both on the suction and pressure surface.

The r/Rs mark the spanwise position of a profile in a blade, which equals 1 at the blade tip and 0 at the blade root. In Figs. 13–14, the pressure coefficient on the profiles of $r/R = 0.5$–0.95 is shown separately. It is seen in both surge and pitch cases that the pressure distribution varies with the position of the profile. The closer to the tip, the lines from different times at each profile get compact, indicating a decrease in the coefficients. But with significantly higher velocities, whose influence has been revealed in Eq. (13), the values of pressure are, in fact, growing, which can also be seen from Fig. 15. Generally, the pressure coefficient for the pitch case is more sensitive to the profile position than that for the surge case. Under the influence of the distance between profiles and the pitch center, the value curves at $0T$ and $1T$ in the pitch case no longer overlap, compared with those of the surge case. The pressure of $0T$ is larger at $x/c = 0$~0.6 on the tip ($r/R = 0.9$ and

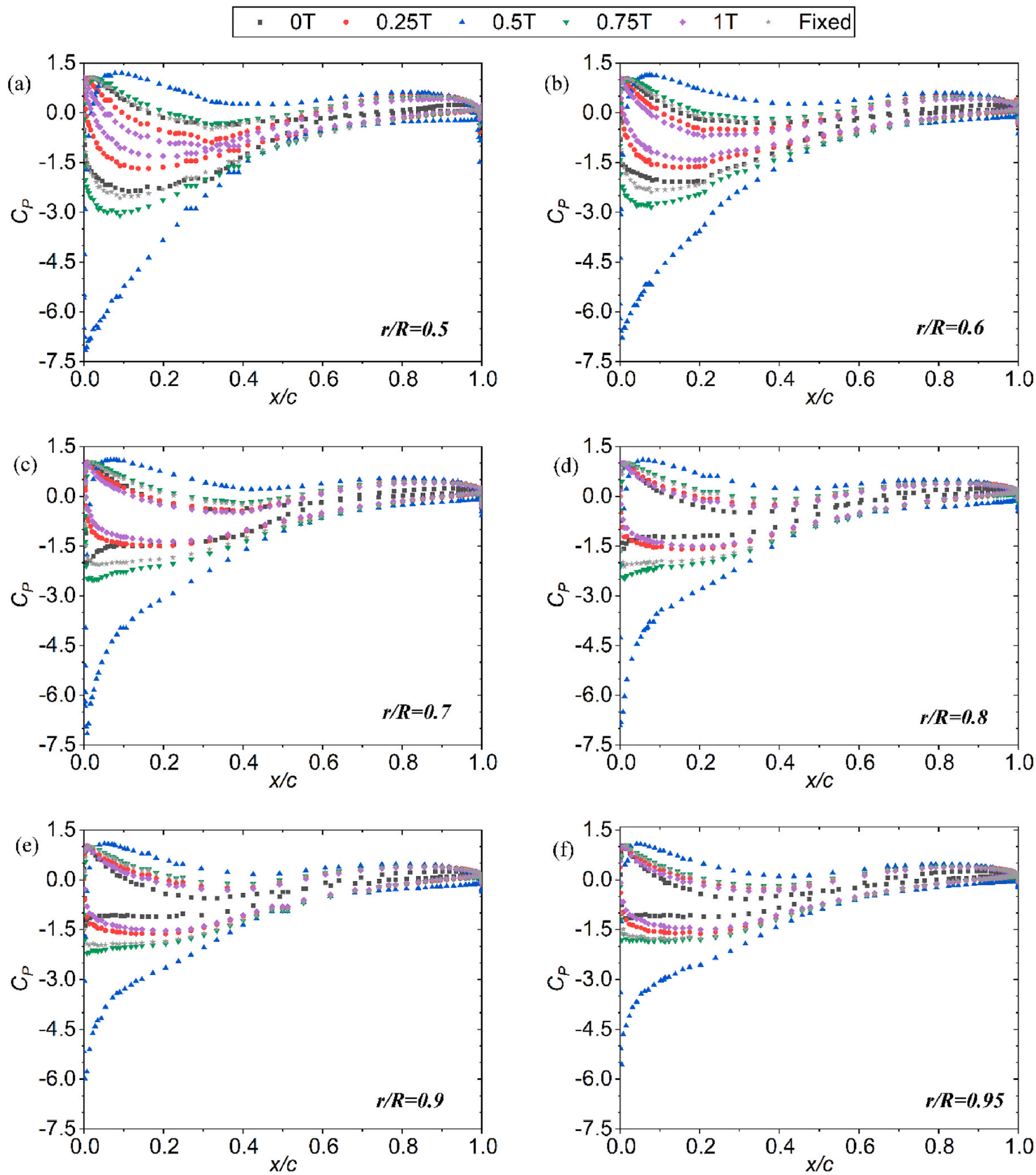

Fig. 14. Pressure coefficients under pitch motion at the spanwise position of r/R = 0.5–0.95 within one period
(a) r/R = 0.5 (b) r/R = 0.6 (c) r/R = 0.7 (d) r/R = 0.8 (e) r/R = 0.9 (f) r/R = 0.95.

0.95) and at the leading edge ($x/c = 0$) from $r/R = 0.5$ to $r/R = 0.8$. Also, the curves in the pitch case near the leading edge present variable tendencies at different spanwise positions. It is clearly shown in Fig. 14 (a) that the curves undergo a first-increase-then-decrease tendency, while as it draws closer to the blade tip, the tendency becomes flat-decrease. Such complication results from the combined influence of profile position and pitch angle, which have also been demonstrated earlier in flow patterns.

As depicted in Fig. 15, the pressure distribution of the rotor blades in the surge, pitch, and fixed cases presents similar patterns. The maximum and minimum pressure of the pitch and surge cases are numerically larger than those of the fixed case. Notably, in the pressure contour of a blade at 0.5*T*, the pressure surface is filled with positive pressure regions. The negative pressure plays a vital role in the suction surface and on the leading edge, which is most evident in the pitch case. Consequently, the pressure value drops sharply from the positive surface to the suction surface, leaving a clear diving line on the leading edge in the contour, which calls for an intensified consideration of structural integrity. Also, it is evident in the contour that the area closer to the tip sustains higher pressure, both on the suction and pressure surfaces.

3.1.3. Overall performance

The performance values of the offshore wind turbine are calculated from Eq. (10)~(12) in Section 2.5.1. The thrust, torque, and aerodynamic power curves for the pitch and surge cases are compared with the fixed ones in Fig. 16. The values of surge and pitch cases are plotted in solid red lines, while the ones for the fixed case are in black dotted lines.

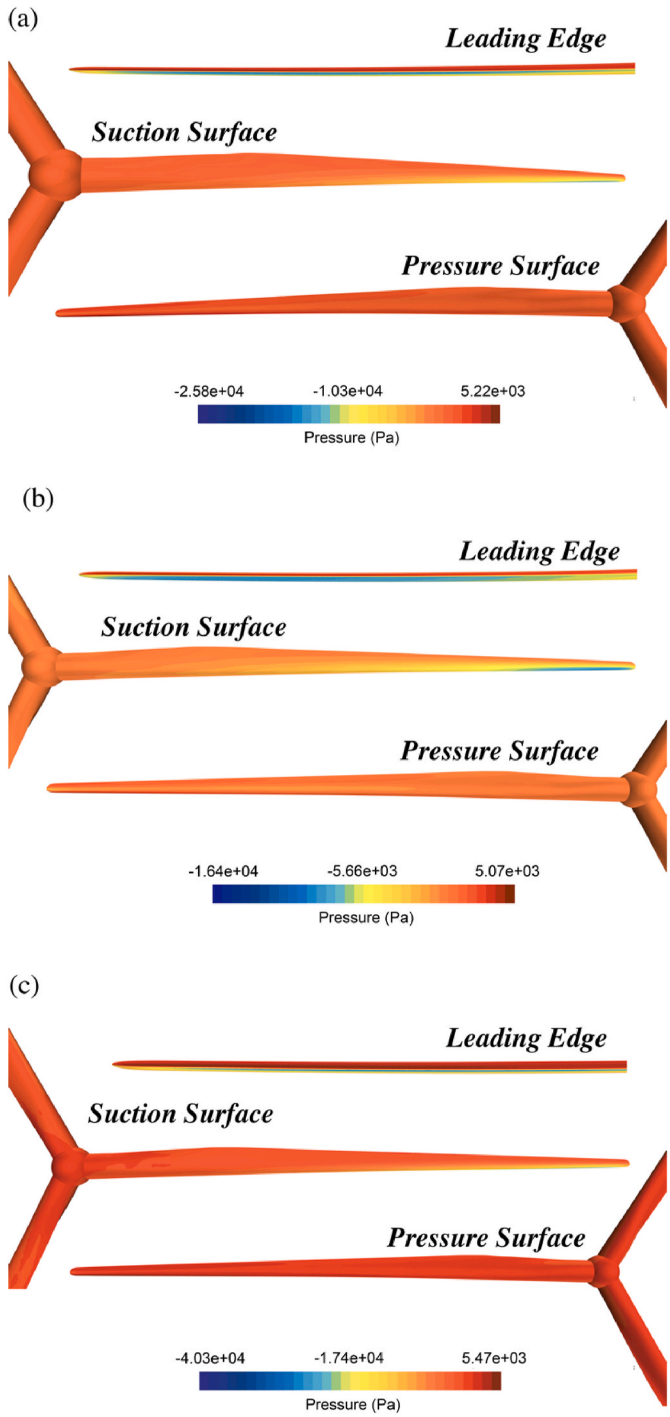

Fig. 15. Pressure distribution of a blade at $0.5T$ for surge, pitch, and fixed cases (a) Surge; (b) Pitch; (c) Fixed.

From Fig. 16(a)(c)(e), it is indicated that for the offshore wind turbine with a simple-harmonic surge motion, the thrust, torque, and power values oscillate in a sinusoidal behavior of the surge period 4s. This can also be observed from the pitch motion cases in Fig. 16(b)(d)(f). The trough and crest in each performance curve correspond to the largest and lowest relative velocity, respectively, toward upwind and downwind positions. Most importantly, it is shown that the aerodynamic power, which is the product of torque and rotational speed, fluctuates severely in floating situations. This rapid variation of performance values gives two essential pieces of knowledge. Firstly, due to the higher relative velocities between the rotor and the wind, as indicated in Fig. 11, the maximum performance values under surge and pitch motion are significantly greater than those under fixed cases. For example, the maximum values of the aerodynamic power reach 2.60 and 4.25 times the rated power in the current pitch and surge cases, respectively. Secondly, the minimum values of the torque and power are close to zero, which would lead to significant instability in the energy production and possibly the sudden halt of the wind turbine gearbox. This is especially true for the pitching rotor, which forms a more extended period around zero in its torque and power curves. Comparing the pitch and surge cases, it can be found that the amplitudes of all performance curves of the pitch case are larger than those of the surge case. This can be attributed to different amplitudes of platform motions since previous studies have already reported the dependence of aerodynamic performances on amplitudes and periods of motions. (Chen et al., 2021; Fang et al., 2021). Overall, it is suggested that excessive pitch and surge motions would jeopardize the regular operation of the offshore wind turbine. Thus, extreme wind and wave conditions should be strictly forbidden. Moreover, the periodic fluctuation of thrust value indicates cyclic loads sustained by FHAWTs and potential structural fatigue. To reveal the resultant variations of tip deformation and stress distribution, the structure solver conducts further structural analysis.

3.2. Structural response

3.2.1. Blade tip deformation

Even though there has been extensive investigation on the unsteady aerodynamic characteristics of HAWTs, the structural responses still await closer revelation. With the one-way FSI approach proposed, this paper extends the study of FHAWTs into the structural criterion. Mapping the aerodynamic pressure to the FE model, the deformation and stress distribution of a wind turbine blade is calculated by Nastran.

The deformation of a single blade in the fixed case is depicted in Fig. 17. It is clear that the deformation is much more violent on the tip than on the root, which results from thinner plies of materials on the tip, as is shown in Fig. 7. The blade tip deformations within one period are compared among the fixed, surge, and pitch cases in Fig. 18, in which the positive deformation is along the positive x-direction (the downwind direction), as Fig. 1 (a) shows. Concerning the surge case, the blade tip deformation undergoes periodic oscillation. During one period, the maximum value occurs at $0.5T$ when the rotor passes through the equilibrium position with the highest relative velocity. The deformation is decreased compared to the fixed case as the rotor moves with the lowest relative velocity ($0T$ and $1T$). At $0.25T$ (upwind-most position), it is slightly smaller than that in the fixed case while being larger at $0.75T$ (downwind-most position). These correspond to the varying trend of the aerodynamic pressure, demonstrating that the aerodynamic pressure plays a significant role in the surge-induced blade tip deformation. Whereas, for the cases in the pitch motion, the tip deformation at the moments of $0T$ and $0.25T$ is altered to the upwind direction. The former is mainly attributed to the negative values in the aerodynamic pressure and relative velocity, as shown in Figs. 11 and 14. Similarly, corresponding to the periodic changes in aerodynamic pressure, the blade has a greater deformation than the fixed case at $0.5T$ and a smaller one at $1T$. While for the latter, the alteration could be due to the inertial loads induced by the pitch motion, as indicated in Eq. (6). This also introduces the largest deformation within the period at $0.75T$.

Specifically, the largest tip deformation in the surge and pitch cases is 1.41 times and 1.73 times that without motions, respectively. This demonstrates that the aerodynamic pressure and the inertial loads induced by pitch and surge motions significantly amplify blade tip deformation. This should alarm the designers, as emphasized in the Introduction, since the structural design of slender blades is primarily driven by deformation and tower clearance. As shown in Eq. (14), the designed tower clearance C of the 15 MW RWT is calculated as 36.127 m, taking into consideration the prebend (L_{PB}, with a unit of m), precone angle (θ_{PC}, with a unit of rad), shaft tilt angle (θ_{ST}, with a unit of rad) and

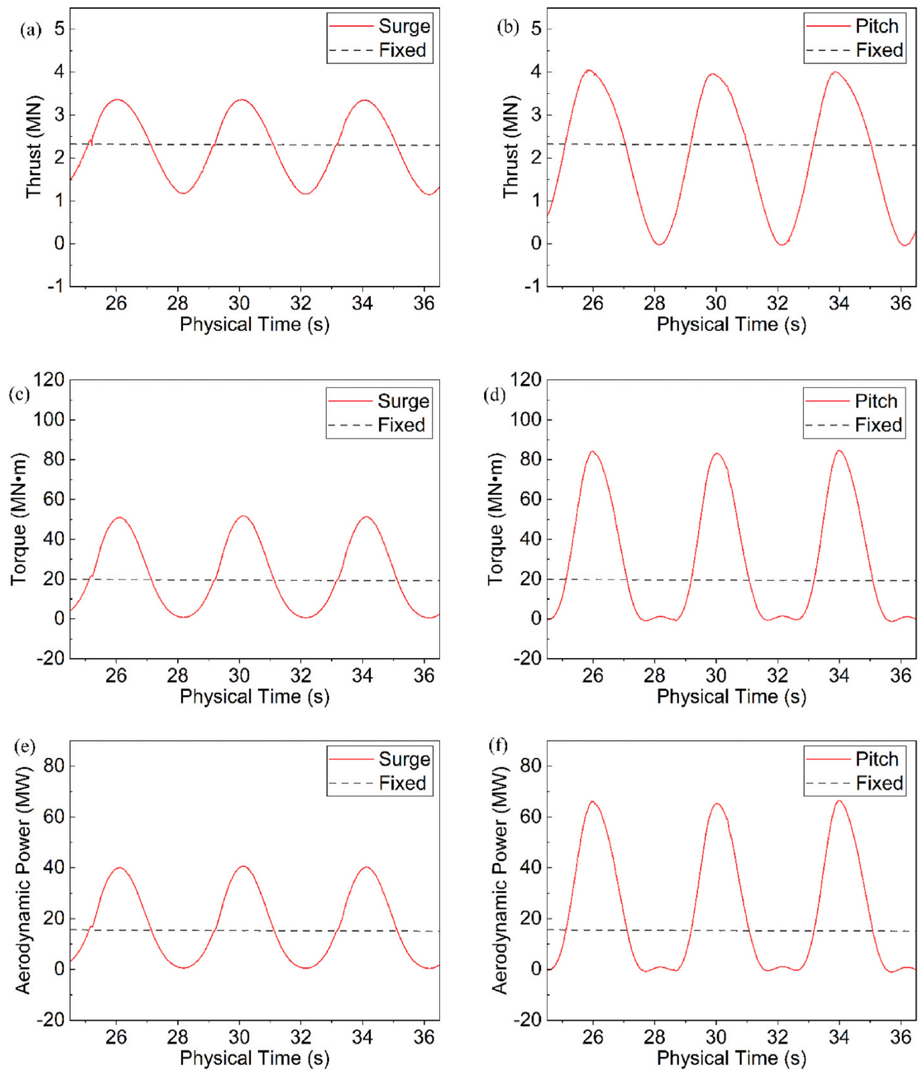

Fig. 16. Overall performance in surge, pitch, and fixed cases (a)~(b) Thrust values for pitch and surge cases; (c)~(d) Torque values for pitch and surge cases; (e)~(f) Aerodynamic power values for pitch and surge cases.

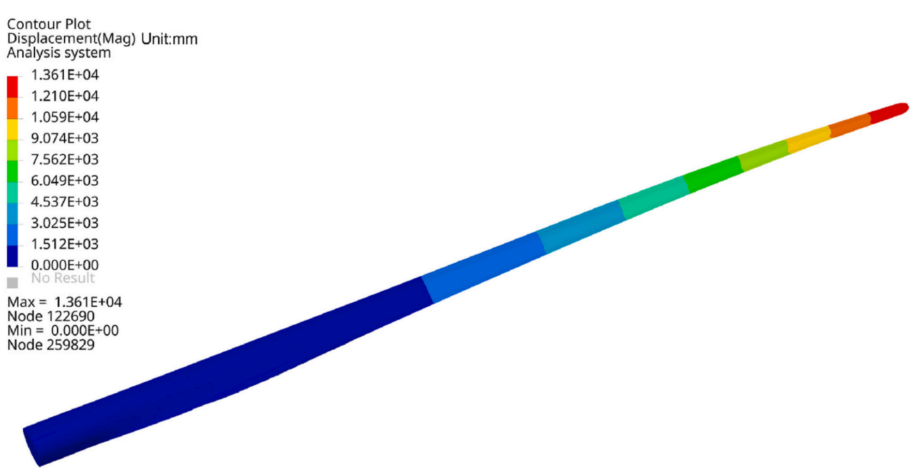

Fig. 17. Deformation of a single blade in the fixed case.

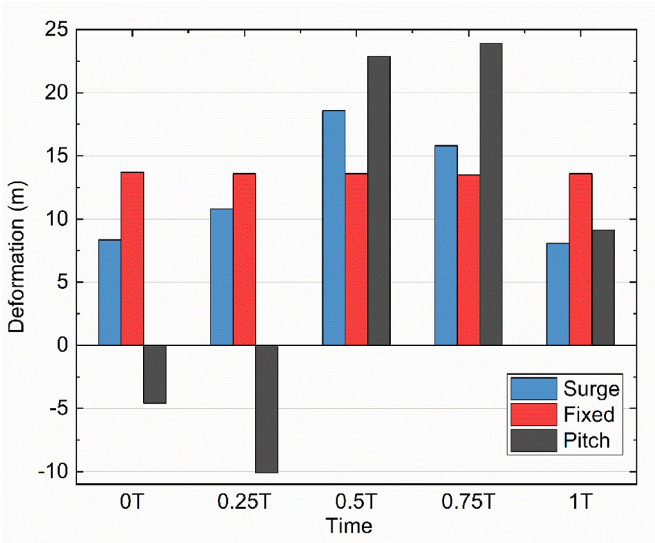

Fig. 18. Blade tip deformations of surge, pitch, and fixed cases within one period.

overhang (L_{OH}, with a unit of m).

$$C = L_{OH} + \sqrt{L_{PB}^2 + R^2}\cdot\sin\left[\tan^{-1}\left(\frac{L_{PB}}{R}\right) + \theta_{PC} + \theta_{ST}\right] \tag{14}$$

Hence, in this study, the amplified deformation values in pitch and surge cases are within this tolerance. However, as a severe sea state may occur with greater wind velocity and stronger wave excitation, such underestimated wind speed provides inadequate assurance. Future work must be probed into the amplification of blade tip deformation to examine the safety of FHAWTs' slender blades.

3.2.2. Stress distribution

The blade stress distribution of the surge, pitch, and fixed cases is presented in Fig. 19. It is revealed that the spar caps of a blade sustain relatively large stress compared to other regions in all situations. Regions on the blade shell without spar cap resist minor stress, especially on the leading and trailing edges. As shown in Fig. 19 (a), the maximum von Mises stress of the blade shell in the fixed case appears at approximately 1/3 spanwise from the blade root, indicating the most prone part to failure in engineering practice. The thickness of spar caps in this area has been designed with extra thickness, as is revealed in Fig. 7, which contributes to the stiffness and structural integrity of the blade. In the spanwise direction, the stress reduces smoothly as it draws away from this region, and the blade's root resists relatively large stress while that on the tip is quite small. In addition, around 1/3 spanwise from the blade tip, the shear webs possess higher von Mises stress. This should be paid special attention to since the materials' thickness and the width of shear webs are quite small to resist possible fractures.

Fig. 18 (b)~(c) shows that the overall stress distributions of the surge cases remain similar patterns as the fixed at different moments, while the values of the stress variate periodically. The maxima of von Mises stress experiences a higher value than the fixed case as the rotor passes through the equilibrium position, moving toward the upward position (0.5 T), and lower values at 0 T and 1 T. These fluctuations are consistent with the pressure distribution, performance parameters, and deformations, as shown in Figs. 15–18. With the influence of inertial surge inertial loads, the stress at 0.75*T* (downwind-most position) is larger than 0.25*T* (upwind-most position). However, in the pitch case, the inertial loads have a more evident influence since the largest stress appears at 0.75 T rather than 0.5 T, and the von Mises stress maxima are closer to the root at 0.25*T* and 0.75*T*. The combined effects of aerodynamic pressure and the inertial loads yielding with the platform motions raise

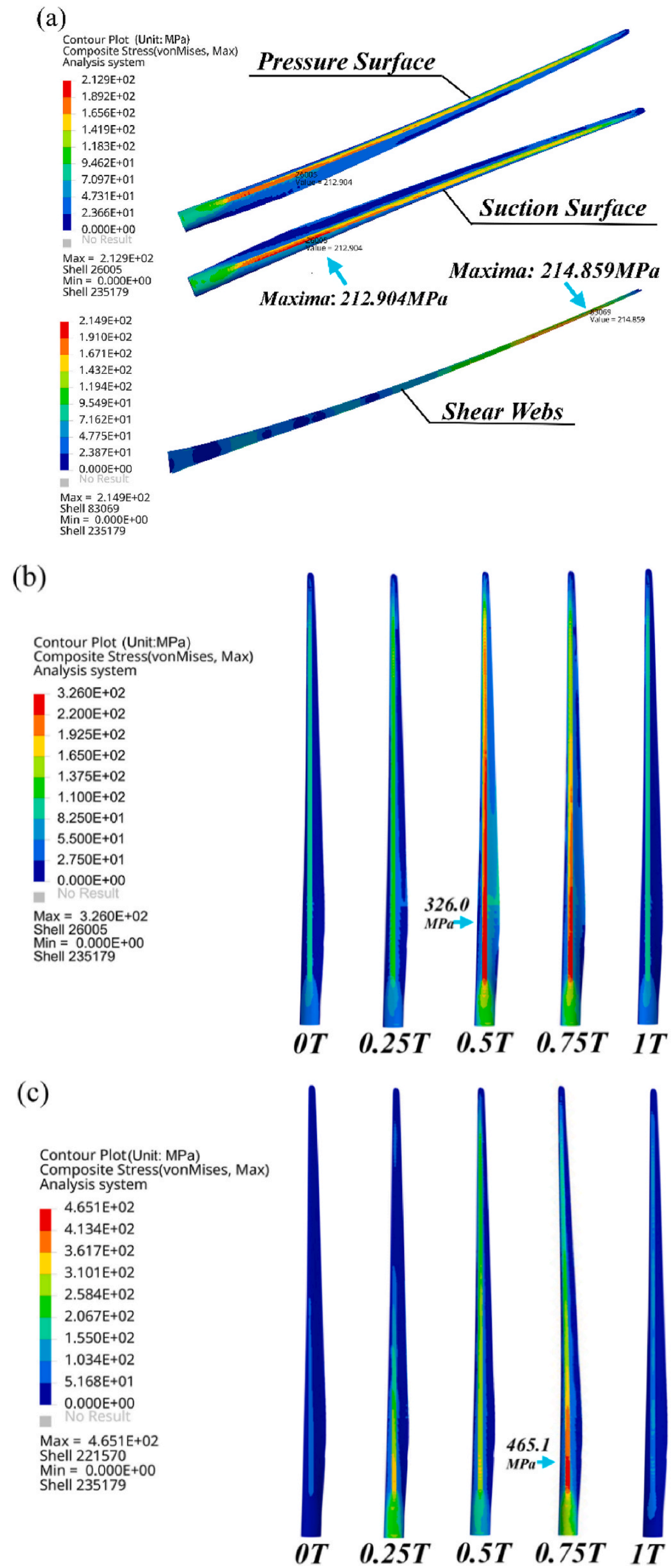

Fig. 19. Stress distribution of a blade for surge, pitch, and fixed cases.

periodic variations in the stresses of blades. The maximum von Mises stresses on the blade shell for the surge and pitch cases are 1.52 and 2.16 times that of the fixed case. This verifies again that the extra motions of the offshore floating wind turbine would raise a higher risk of fatigue on the blade structures.

To present the influence of the composite structure of the blade,

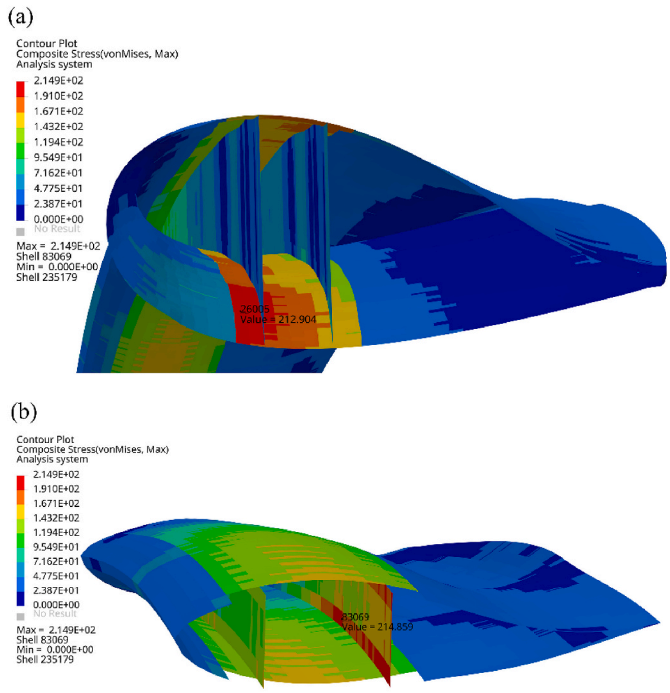

Fig. 20. Stress distribution near the positions of maxima on the blade shell and the shear webs (a) The maxima on blade shell; (b) The maxima on shear webs.

Fig. 20 sheds more light on the stress distribution of the regions near the maxima values on the blade shell and the shear webs. It is shown in Fig. 20 (a) that the maxima appear on blade shell with spear caps, and the stress drops severely from regions with spar caps to regions without, indicating potential safety pitfalls. Fig. 20 (b) suggests that the edges of spar caps sustain higher stresses. A more in-depth investigation should be carried out in the future targeting such regions since the linking between blade shell and shear webs are a crucial aspect of a blade's safety.

4. Conclusions

In this study, the aerodynamic analysis of the floating 15 MW wind turbine was carried out utilizing the CFD-FEA one-way FSI approach at full scale. Taking into account the complex outer shape and the detailed composite lay-ups, the CFD and FE models were established separately, and the aerodynamic pressure acquired from the CFD simulation was mapped to the FE elements as boundary conditions. Validation and resolution studies have been performed to validate the FSI model. In this research, the influence of platform motions on the blade's aerodynamic and structural characteristics is discussed in detail. The main conclusions are drawn as follows.

i. Due to the platform motions, the relative velocities between the blade and the wind field alter periodically, resulting in an evidently disturbed wind velocity field and distorted wake flows around the blades. The maximum pressure is significantly amplified for the pitch and surge cases compared with that of the fixed case.
ii. The surge and pitch motions draw periodic curves in performance values with the same period as the motions, revealing that the motions amplify the maximum thrust, torque, and aerodynamic power. In this study scope, the maximum values of the aerodynamic power in pitch and surge cases reach 2.60 and 4.25 times the rated power, respectively. The minimum values of the torque and power are close to zero, which would lead to significant instability in the energy production and possibly the sudden halt of the wind turbine gearbox.
iii. The blade deformation of a single blade is much more violent on the tip than on the root. Within one period, the maximum tip deformations in the surge and pitch cases are 1.41 times and 1.73 times that without motions, respectively. The amplification should alarm the designers that severe sea state should be considered in further investigations to confirm its safety.
iv. The stress distribution shows that the spar caps of a blade sustain relatively large stress compared to other regions. The maximum von Mises stress of the blade shell appears at approximately 1/3 spanwise from the blade root. The shear webs around 1/3 spanwise from the blade tip possess high von Mises stress. In surge and pitch cases, the maxima on the blade shell for the surge and pitch cases are 1.52 and 2.16 times that of the fixed case. This indicates potential fatigue and failures, which should be paid special attention to in design.

In addition, it is important to note that the validity of the one-way FSI model established in this study is not verified by a two-way model due to limited resources and time and such a comparative study is necessary to be performed in the future.

CRediT authorship contribution statement

Yuchen Zhang: Writing – original draft, Writing – review & editing, Formal analysis, Investigation. **Yue Song:** Writing – review & editing, Supervision. **Cong Shen:** Investigation, Data curation. **Nian-Zhong Chen:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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